Theorem. *In an integral domain* A, *for every polynomial* P *in the variable* x *with coefficients in* A, $P \in \mathsf{A}[x]$, *and every* $r \in \mathsf{A}$, *there exist unique* $\mathcal{Q} \in \mathsf{A}[x]$ *and* $\mathcal{R} \in \mathsf{A}$ *such that*

$$P(x) = (x - r)\,\mathcal{Q}(x) + \mathcal{R},$$

where

$$\mathcal{R} = P(r).$$

○ 1. The existence

Let $P(x)$ be a polynomial in $\mathsf{A}[x]$ of degree n,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^{n} a_i x^i,$$

where each coefficient $a_i \in \mathsf{A}$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^{n} a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i = 0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned} \sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i &= (a_0 - a_0) + \sum_{i=1}^{n} \Big(a_i x^i - a_i r^i \Big) \\ &= \sum_{i=1}^{n} a_i \big(x^i - r^i \big). \end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i-th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

Lemma. *For any integer $n \geq 1$, the difference of two n-th powers, $x^n - c^n$, can be factored by $(x-c)$. That is, there exists a polynomial $S_{n-1}(x,c)$ such that*

$$x^n - c^n = (x-c)\,S_{n-1}(x,c).$$

○ (by mathematical induction on n)

The **base case** $n=1$ holds since

$$x^1 - c^1 = (x-c)\,(1)\,,$$

where the constant $1 = S_0(x,c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $S_{k-1}(x,c)$ such that

$$x^k - c^k = (x-c)\,S_{k-1}(x,c).$$

Induction Step. To show that the lemma holds for $k+1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^k x$ and grouping as

$$x^{k+1} - xc^k + c^k x - c^{k+1} = \big(x^{k+1} - xc^k\big) + \big(c^k x - c^{k+1}\big).$$

Factoring out the common components from each group gives

$$x\big(x^k - c^k\big) + c^k(x-c).$$

Applying the inductive hypothesis to substitute $(x-c)\,S_{k-1}(x,c)$ for $\big(x^k - c^k\big)$ yields

$$x(x-c)\,S_{k-1}(x,c) + c^k(x-c).$$

Factoring out the common multiplier $(x-c)$ results in

$$(x-c)\Big(x\,S_{k-1}(x,c) + c^k\Big),$$

where $x\,S_{k-1}(x,c) + c^k = S_k(x,c)$ is a polynomial — since $S_{k-1}(x,c)$ is a polynomial by the inductive hypothesis, and polynomial rings are closed under multiplication and addition.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $S(x,c)$ polynomials are

$$\begin{aligned}
1 &&= S_0\\
x(1) + c^1 = x + c &&= S_1\\
x(x+c) + c^2 = x^2 + xc + c^2 &&= S_2\\
x\big(x^2 + xc + c^2\big) + c^3 = x^3 + x^2c + xc^2 + c^3 &&= S_3\\
x\big(x^3 + x^2c + xc^2 + c^3\big) + c^4 = x^4 + x^3c + x^2c^2 + xc^3 + c^4 &&= S_4\\
\cdots\\
x\big(x^{n-2} + x^{n-3}c + \cdots + xc^{n-3} + c^{n-2}\big) + c^{n-1} = x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1} &&= S_{n-1}\\
x\big(x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1}\big) + c^{n} = x^{n} + x^{n-1}c + \cdots + xc^{n-1} + c^{n} &&= S_{n}
\end{aligned}$$

In more compact notation with the summation symbol

$$S_n(x,c) = x^n + x^{n-1}c + \cdots + xc^{n-1} + c^n = \sum_{i=0}^{n} x^{n-i}c^i$$

and

$$x\,S_{n-1}(x,c) + c^n = x\sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^{n} x^{n-i}c^i = S_n(x,c)\,.$$

The exact factorization is $$x^n - c^n = (x-c)\sum_{i=0}^{n-1} x^{n-1-i}c^i.$$

Several interesting properties characterize $S_n(x,c) = \sum_{i=0}^{n} x^{n-i}c^i$ as a *complete homogeneous symmetric polynomial* in two variables :

- ✓ it is *symmetric*, since $S_n(x,c) = \sum_{i=0}^{n} x^{n-i}c^i = \sum_{i=0}^{n} c^{n-i}x^i = S_n(c,x)$,
- ✓ it is *homogeneous* of degree n, because the degree of each monomial $x^{n-i}c^i$ is exactly $n-i+i=n$ for all i,
- ✓ it is *complete* in the sense that it contains every possible n-th degree monomial in the variables x and c exactly once, each with a coefficient of 1.

The symmetry of $S_n(x,c)$ is already hinted at in the induction step. The proof inserts the zero $0 = -xc^k + c^kx$, but one may equally insert the zero $0 = -x^kc + cx^k$. Then $x^{k+1} - x^kc + cx^k - c^{k+1} = x^k(x-c) + c\big(x^k - c^k\big)$ and the resulting recurrence will be

$$S_k(x,c) = x^k + c\,S_{k-1}(x,c)\,.$$

The recurrences $x^k + c\,S_{k-1}(x,c)$ and $x\,S_{k-1}(x,c) + c^k$ are dual under the swap $x \leftrightarrow c$.

Returning to $$P(x) - P(r) = \sum_{i=1}^{n} a_i\big(x^i - r^i\big),$$

the lemma yields*

$$x^i - r^i = (x-r)\,S_{i-1}(x,r)$$

for every $i \geq 1$.

* From a deeper algebraic viewpoint, $S_{i-1}(x,c)$ from the lemma belongs to the polynomial ring $\mathsf{A}[x,c]$. Replacing the indeterminate c by the element $r \in \mathsf{A}$ is the evaluation homomorphism $\mathsf{A}[x,c] \to \mathsf{A}[x]$. Thus $S_{i-1}(x,r)$ is a polynomial in $\mathsf{A}[x]$.

Substituting these factorizations term-by-term gives

$$P(x)-P(r)=\sum_{i=1}^{n} a_i\bigl(x-r\bigr)S_{i-1}(x,r)=\bigl(x-r\bigr)\sum_{i=1}^{n} a_i\,S_{i-1}(x,r).$$

Since each $S_{i-1}(x,r)$ is a polynomial in $\mathsf{A}[x]$ and polynomial sums are polynomials, the sum $\sum_{i=1}^{n} a_i\,S_{i-1}(x,r)$ belongs to the same polynomial ring $\mathsf{A}[x]$. Define

$$\sum_{i=1}^{n} a_i\,S_{i-1}(x,r)=\mathcal{Q}(x)\in\mathsf{A}[x]\,.$$

Hence $\qquad P(x)-P(r)=\bigl(x-r\bigr)\,\mathcal{Q}(x)$

or equivalently $\qquad P(x)=\bigl(x-r\bigr)\,\mathcal{Q}(x)+P(r)$

— the latter exhibits the intended representation with $\mathcal{R}=P(r)\in\mathsf{A}$. Ergo, such $\mathcal{Q}\in\mathsf{A}[x]$ and $\mathcal{R}\in\mathsf{A}$ exist. ●

It remains to show that $\mathcal{Q}(x)$ and $\mathcal{R}$ are unique.

○ 2. The uniqueness

Assume that there exist polynomials $Q', Q''\in\mathsf{A}[x]$ and constants $R', R''\in\mathsf{A}$ such that

$$P(x)=(x-r)\,Q'(x)+R' \quad\text{and}\quad P(x)=(x-r)\,Q''(x)+R''.$$

Subtracting the second representation from the first gives

$$(x-r)\,Q'(x)+R'-(x-r)\,Q''(x)-R''=0\,,$$

which simplifies to

$$(x-r)\bigl(Q'(x)-Q''(x)\bigr)=R''-R'.$$

Evaluating this polynomial identity at $x=r$, the factor $x-r$ becomes $r-r=0$, so the left-hand side vanishes and reveals that the constant difference is zero :

$$0=R''-R',$$

that is

$$R'=R''.$$

Returning to the polynomial identity before evaluation and putting $R'=R''$ into it yields

$$(x-r)\bigl(Q'(x)-Q''(x)\bigr)=0\,.$$

Because A is an integral domain, the polynomial ring $\mathsf{A}[x]$ is an integral domain as well and therefore has no zero divisors. Moreover, $(x-r)$ is a nonzero polynomial in $\mathsf{A}[x]$. Since its product with $Q'(x)-Q''(x)$ is the zero polynomial, the absence of zero divisors in $\mathsf{A}[x]$ implies that

$$Q'(x)-Q''(x)=0\,,$$

giving

$$Q'(x)=Q''(x)\,.$$

Thus

$$Q'=Q'' \quad \text{and} \quad R'=R'',$$

showing that the intended representation

$$P(x)=(x-r)\,\mathcal{Q}(x)+\mathcal{R}$$

is unique in the choice of $\mathcal{Q}$ and $\mathcal{R}$. ●